

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping customer-facing AI systems end-to-end — multi-provider LLM abstractions, schema-validated agent handoffs, FastAPI/Flask packaging, and HIPAA-conscious governance. Track record of measurable customer outcomes: ~25% agent stability gain, ~35% retrieval precision improvement, ~30% post-deployment defect reduction. Founder of E-Farming marketplace (onboarded 80–120 active users phase one) and solo shipper of **Manga Lens on the Chrome Web Store** (4 AI vision providers behind one abstraction, 29 sites, customer-debug across DOM structures). Comfortable reading and debugging unfamiliar codebases in Python and JavaScript, and communicating complex AI to technical and non-technical stakeholders.

CORE COMPETENCIES

Multi-Provider LLM Integration • Customer-Facing Solution Design • Production AI Debugging • Python / JavaScript / TypeScript • FastAPI / Flask / Docker • RAG & Evaluation Frameworks • Structured Prompt Transformation • Chrome MV3 / Service Workers (event-driven) • HIPAA-Conscious Governance

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 – Present

Jr. AI/ML Engineer Trainee

USA · Healthcare Technology, AI Consulting, SaaS

- Packaged inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects ~**30%**.
- Shipped **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, grounding rules, and audit trails; improved agent response stability by ~**25%**.
- Built **Retrieval-Augmented Generation (RAG) systems** with recursive semantic chunking and transformer embeddings; improved contextual retrieval precision ~**35%** and held retrieval-grounded response alignment **above 90%** in evaluation.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall **15–20%** on high-risk categories.
- Translated field learnings into stakeholder-facing system-limitation docs, evaluation audit trails, and HIPAA-conscious data lineage; analogous discipline to DPAs/MSAs/security review documentation.

Energy Solutions International (Emerson)

Jun 2022 – Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India · Oil & Gas, Enterprise ERP, Industrial Automation

- Built **CI/CD pipelines with Jenkins** for schema updates, version-controlled SQL scripts, and stored procedure deployments; reduced deployment errors >**30%** and improved release cycle time ~**35–40%**.
- Designed **Grafana + SQL Server DMV dashboards** for long-running queries and deadlocks; cut incident recurrence ~**25%** and improved root-cause resolution speed ~**30%**.
- Tuned T-SQL stored procedures powering Order-to-Cash workflows; reduced query execution time ~**20%** and data retrieval latency ~**25%**.

Suvidha Foundation

Jan 2022 – Mar 2022

Machine Learning Engineer

Hyderabad, India · Nonprofit Tech, Video / Audio Analytics, Applied NLP

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; transcript preprocessing, hierarchical summarization, and timestamp-aligned clip extraction reduced manual review time by **60–70%** and reached ~**85% highlight alignment** with human curation.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~**30%** and raised contextual answer accuracy >**85%**.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

SELECTED PROJECTS

Manga Lens — Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack browser extension built independently — Manifest V3, content scripts, service workers, and *captureVisibleTab* for panel capture. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind one abstraction. Multi-section capture handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. **Per-domain selector configs for 29 manga/webtoon sites**, 7-day translation cache, privacy policy, narrowed host permissions, WebP/JPEG provider routing.

Dream Decoder — AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams and synthesizes dreamscapes via coordinated multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Introduced **intermediate structured prompt transformation layers** — improved contextual alignment **~30%** and first-pass image success rate **~25–30%** over naive direct-prompt orchestration.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents**. RAG-grounded CPT/ICD validation; approximate-nearest-neighbor similarity search for duplicates; explainable risk scoring with audit-ready reasoning traces.

Video Summarization & Highlight Generation System (Suvidha Foundation)

Transformer-based hierarchical abstractive summarization on **5,000+ recorded sessions**; transcript preprocessing and timestamp-aligned clip extraction. **60–70% reduction in manual review time**, **~85% highlight alignment** with human curation, Flask-based API for non-technical users.

E-Farming Digital Marketplace (Founder & Full-Stack Developer)

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. **Onboarded 80–120 active users in phase one**; cart, reviews, and community blogging features to drive repeat engagement and expansion.

EDUCATION

Master's Degree - Kent State University, Ohio, USA Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT - US Work Authorization (now) STEM extension eligible

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript, TypeScript, FastAPI, Flask, React, Node tooling (npm), SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

LLM & GenAI: Multi-provider LLM integration, agentic workflows, RAG pipelines, evaluation frameworks, structured prompt transformation, grounding, structured outputs, embeddings, vector search

Customer-Facing & Debugging: Reading and debugging unfamiliar codebases, multi-provider abstraction design, per-domain config debugging, structured logging, stakeholder communication, system-limitation docs

ML & Multimodal: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, YOLOv8, OpenCV, ControlNet, Stable Diffusion, Hugging Face Diffusers

Infra & DevOps: Docker, Jenkins, REST/JSON APIs, Chrome MV3 service workers (event-driven async), CI/CD, observability/logging, cloud-ready deployment

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Customer-Facing AI Delivery, Workflow Automation, Audio/Video Analytics